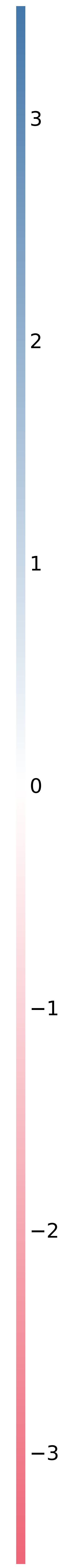
Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:



<think>
The de Broglie wavelength of a particle is given by the equation:

Î» = h / p

where Î» is the de Broglie wavelength, h is the Planck's constant (6.626 ÃĹ 10^-34 J s), and p is the momentum of the particle.

The kinetic energy of an electron is given by:

KE = (1/2)mv^2

where KE is the kinetic energy, m is the mass of the electron (9.11 ÃĹ 10^-31 kg), and v is the velocity of the electron.

We can find the velocity of the electron using the given kinetic energy:

KE = (1/2)mv^2
1 eV = 1.6 ÃĹ 10^-19 J
(1.6 ÃĹ 10^-19 J) = (1/2)(9.11 ÃĹ 10^-31 kg)v^2
v^2 = 2(1.6 ÃĹ 10^-19 J) / (9.11 ÃĹ 10^-31 kg)
v^2 = 3.51 ÃĹ 10^11 m^2/s^2
v = âĪļ(3.51 ÃĹ 10^11 m^2/s^2) = 5.92 ÃĹ 10^5 m/s

Now we can find the momentum of the electron:

p = mv
p = (9.11 ÃĹ 10^-31 kg)(5.92 ÃĹ 10^5 m/s)
p = 5.39 ÃĹ 10^-26 kg m/s

Now we can find the de Broglie wavelength:

Î» = h / p
Î» = (6.626 ÃĹ 10^-34 J s) / (5.39 ÃĹ 10^-26 kg m/s)
Î» = 1.23 ÃĹ 10^-7 m
Î» = 1.23 ÃĹ 10^-7 cm

Therefore, the de Broglie wavelength is 1.23 ÃĹ 10^-7 cm
</think><answer> G. 1.23 ÃĹ 10^-7 cm</answer>